



Fig. 2: Indian PCB market at a glance

Obstacles for growth

- Inefficient supply chain for raw materials resulting in high inventory storage costs
- Unfair level playing due to the high cost of finance with respect to other countries
- Inverted tax structure
- Logistics inefficiencies and infrastructural bottlenecks resulting in greater turnaround time and costs
- Conflict of interest among various sectors within the electronics industry, from components to finished products

Some leading PCB Manufacturers in India

- AT&S India Pvt Ltd
- Ascent Circuits Pvt Ltd
- BLG Electronics Ltd
- CIPSA-TEC India Pvt Ltd
- Circuits Systems India Ltd/PCB Power
- Epitome Components Ltd
- Fine-Line Circuits Ltd
- Garg Electronics Pvt Ltd
- Genus Electrotech Ltd
- Hi-Q Electronics Pvt Ltd
- Meena Circuits Pvt Ltd
- Micropack Ltd
- Shogini Technoarts Pvt Ltd
- Sulakshana Circuits Ltd
- Vintek Circuits (India) Pvt Ltd

New Delhi to Chennai and Gujarat to West Bengal. Because of this, inland transportation costs are more than importing costs from China.

Satish Athavale, managing director, Shogini Technoarts, highlights that, "Presently, output of the Indian PCB industry is less than one per cent of the total global capacity. This is due to unavailability of the latest technology and high capital requirement for capacity expansion." Capital investment cost goes up due to the very high cost of finance in the country for capital investment, as well as

for working capital.

Goel explains that, the very low capital-output ratio in the PCB industry entails need for huge investment and high interest cost. Thus, profitability is an issue that is impacted by the fact that manufacturers are not able to invest in new equipment, most of which is imported. He also mentioned that availability of skilled manpower is also a challenge as the industry is small and does not support an ecosystem for organised skill-development infrastructure.

Infrastructural challenges such as availability of clean and uninterrupted power, water, developed industrial parks and lack of raw material were some of the hindrances to the growth of the PCB industry in the past. According to Muniswamy, "Though things are improving, raw material supply chain is still an issue in the country as most raw materials are still being imported."

Non-availability of critical raw materials increases inventory cost and non-availability of modern manufacturing equipment increases the overall cost of acquisition and maintenance. Vasani says, "There is no domestic supply chain for the raw materials required in the PCB industry. In view of

huge dependence on imports, manufacturers have to keep over three months of inventory, which adds to the cost."

Doshi feels that the large import content of raw materials in a strong dollar situation will significantly impact margins and hinder growth.

These factors contribute towards increasing the price point of the product, which makes these less competitive as compared to their Chinese competitors. Dhoot mentions that, the relevance of manufacturing equipment used today is for a maximum of four years, after which these tend to become redundant.

Vasani also mentioned the need for huge investments associated with the creation and maintenance of effluent treatment facilities as one of the major challenges for this industry.

In this regard, Athavale emphasised the need for support from the government and felt that this could enable India to avail the opportunity that has been created by the shutting down of PCB manufacturing in European Union (EU) due to stringent pollution-control concerns.

While discussing the capability limitations of the Indian PCB industry, Dhoot says, "India does not manufacture PCBs for most high-growth project segments such as mobile handsets, laptops or set-top boxes. We are only able to manufacture PCBs for consumer electronics such as televisions and power electronics products such as UPSes and inverters. Even LCD/LED televisions require high-tech PCBs with a specification of 5mil x 5mil boards. This often results in high rejection of Indian PCBs in this segment due to lack of an inadequately upgraded projection process in the country."

Most Indian PCB companies in the past catered to a small local market consisting primarily of public sector companies and therefore never felt the need for continual improvement.

To grow in today's competitive environment, however, PCB companies must focus on top-notch quality that could be sold in any market across the globe. Developing a global market for